5.1 Introduction

In **Chapter 4**, we considered discrete random variable—that is, random variables whose set of possible values is either finite or countably infinite. However, there also exist random variables whose set of possible values is uncountable. Two examples are the time that a train arrives at a specified stop and the lifetime of a transistor. Let X be such a random variable. We say that X is a *continuous* † random variable if there exists a nonnegative function f, defined for all real $x \in (-\infty, \infty)$, having the property that for any set B of real numbers, ‡

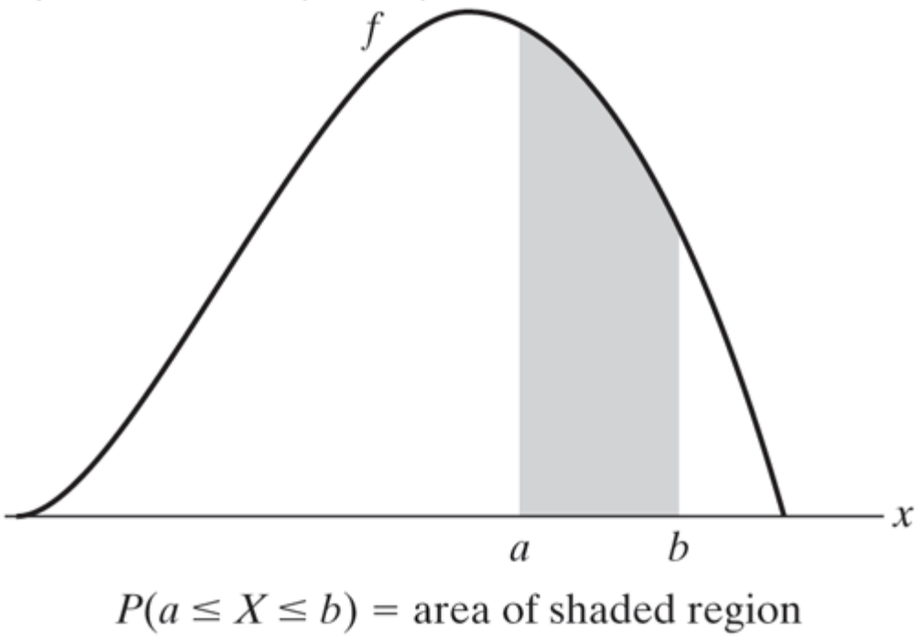
† Sometimes called *absolutely continuous*.

$$P\{X \in B\} = \int_B f(x)dx \tag{1.1}$$

‡ Actually, for technical reasons, **Equation (1.1)** is true only for the *measurable* sets B, which, fortunately, include all sets of practical interest.

The function f is called the *probability density function* of the random variable X. (See **Figure 5.1**.)

Figure 5.1 Probability density function f.



In words, **Equation (1.1)** states that the probability that X will be in B may be obtained by integrating the probability density function over the set B. Since X must assume some value, f must satisfy

$$1 = P\{X \in (-\infty, \infty)\} = \int_{-\infty}^{\infty} f(x)\,dx$$

All probability statements about X can be answered in terms of f. For instance, from **Equation (1.1)**, letting B=[a,b], we obtain

$$P\{a \leq X \leq b\} = \int_a^b f(x)\,dx \tag{1.2}$$

If we let a=b in **Equation (1.2)**, we get

$$P\{X=a\}=\int_a^a f(x)\,dx=0$$

In words, this equation states that the probability that a continuous random variable will assume any fixed value is zero. Hence, for a continuous random variable,

$$P\{X<a\}=P\{X\le a\}=F(a)=\int_{-\infty}^{a} f(x)\,dx$$

Example 1a

Suppose that X is a continuous random variable whose probability density function is given by

$$f(x)=\begin{cases}C(4x-2x^2) & 0<x<2\\ 0 & \text{otherwise}\end{cases}$$

a. What is the value of C?
b. Find $P\{X>1\}$.

Solution

a. Since f is a probability density function, we must have $\int_{-\infty}^{\infty} f(x)\,dx=1$, implying that
$C\int_0^2(4x-2x^2)\,dx=1$

or
$C[2x^2-\frac{2x^3}{3}]\Big|_{x=0}^{x=2}=1$

or
$C=\frac{3}{8}$

Hence,
b. $P\{X>1\}=\int_1^{\infty} f(x)\,dx=\frac{3}{8}\int_1^2(4x-2x^2)dx=\frac{1}{2}$

Example 1b

The amount of time in hours that a computer functions before breaking down is a continuous random variable with probability density function given by

$$f(x)=\begin{cases}\lambda e^{-x/100} & x\ge 0\\ 0 & x<0\end{cases}$$

What is the probability that

a. a computer will function between 50 and 150 hours before breaking down?
b. it will function for fewer than 100 hours?

Solution

a. Since
$1=\int_{-\infty}^{\infty} f(x)\,dx=\lambda\int_0^{\infty} e^{-x/100}\,dx$

we obtain
$1=-\lambda(100)e^{-x/100}\Big|_0^{\infty}=100\lambda \quad \text{or} \quad \lambda=\frac{1}{100}$

Hence, the probability that a computer will function between 50 and 150 hours before breaking

down is given by
$P\{50<X<150\}=\int_{50}^{150}\frac{1}{100}e^{-x/100}\,dx=-e^{-x/100}\big|_{50}^{150}=e^{-1/2}-e^{-3/2}\approx.383$

b. Similarly,
$P\{X<100\}=\int_{0}^{100}\frac{1}{100}e^{-x/100}\,dx=-e^{-x/100}\big|_{0}^{100}=1-e^{-1}\approx.632$

In other words, approximately 63.2 percent of the time, a computer will fail before registering 100 hours of use.

Example 1c

The lifetime in hours of a certain kind of radio tube is a random variable having a probability density function given by

$f(x)=\begin{cases}0 & x\le 100\\ \frac{100}{x^2} & x>100\end{cases}$

What is the probability that exactly 2 of 5 such tubes in a radio set will have to be replaced within the first 150 hours of operation? Assume that the events Ei, i=1,2,3,4,5, that the *i*th such tube will have to be replaced within this time are independent.

Solution

From the statement of the problem, we have

$P(E_i)=\int_{0}^{150}f(x)\,dx=100\int_{100}^{150}x^{-2}\,dx=\frac{1}{3}$

Hence, from the independence of the events Ei, it follows that the desired probability is

$\binom{5}{2}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3=\frac{80}{243}$

The relationship between the cumulative distribution F and the probability density f is expressed by

$F(a)=P\{X\in(-\infty,a]\}=\int_{-\infty}^{a}f(x)\,dx$

Differentiating both sides of the preceding equation yields

$\frac{d}{da}F(a)=f(a)$

That is, the density is the derivative of the cumulative distribution function. A somewhat more intuitive interpretation of the density function may be obtained from **Equation (1.2)** as follows:

$P\{a-\frac{\varepsilon}{2}\le X\le a+\frac{\varepsilon}{2}\}=\int_{a-\varepsilon/2}^{a+\varepsilon/2}f(x)\,dx\approx\varepsilon f(a)$

when ε is small and when f(·) is continuous at x=a. In other words, the probability that X will be contained in an interval of length ε around the point a is approximately εf(a). From this result, we see that f(a) is a measure of how likely it is that the random variable will be near a.

Example 1d

If X is continuous with distribution function FX and density function fX, find the density function of Y=2X.

Solution

We will determine fY in two ways. The first way is to derive, and then differentiate, the distribution function of Y:

$$F_Y(a) = P\{Y \le a\} = P\{2X \le a\} = P\{X \le a/2\} = F_X(a/2)$$

Differentiation gives

$$f_Y(a) = \frac{1}{2} f_X(a/2)$$

Another way to determine f_Y is to note that

$$\varepsilon f_Y(a) \approx P\{a - \frac{\varepsilon}{2} \le Y \le a + \frac{\varepsilon}{2}\} = P\{a - \frac{\varepsilon}{2} \le 2X \le a + \frac{\varepsilon}{2}\} = P\{\frac{a}{2} - \frac{\varepsilon}{4} \le X \le \frac{a}{2} + \frac{\varepsilon}{4}\} \approx \frac{\varepsilon}{2} f_X(a/2)$$

Dividing through by ε gives the same result as before.

5.2 Expectation and Variance of Continuous Random Variables

In **Chapter 4**, we defined the expected value of a discrete random variable X by

$$E[X]=\sum_x xP\{X=x\}$$

If X is a continuous random variable having probability density function f(x), then, because

$$f(x)\,dx \approx P\{x \le X \le x+dx\} \quad \text{for } dx \text{ small}$$

it is easy to see that the analogous definition is to define the expected value of X by

$$E[X]=\int_{-\infty}^{\infty} xf(x)\,dx$$

Example 2a

Find E[X] when the density function of X is

$$f(x)=\begin{cases}2x & \text{if } 0 \le x \le 1\\ 0 & \text{otherwise}\end{cases}$$

Solution

$$E[X]=\int xf(x)\,dx=\int_0^1 2x^2\,dx=\frac{2}{3}$$

Example 2b

The density function of X is given by

$$f(x)=\begin{cases}1 & \text{if } 0 \le x \le 1\\ 0 & \text{otherwise}\end{cases}$$

Find E[eX].

Solution

Let Y=eX. We start by determining FY, the cumulative distribution function of Y. Now, for 1≤x≤e,

$$F_Y(x)=P\{Y \le x\}=P\{eX \le x\}=P\{X \le \log(x)\}=\int_0^{\log(x)} f(y)dy=\log(x)$$

By differentiating FY(x), we can conclude that the probability density function of Y is given by

$$f_Y(x)=\frac{1}{x} \qquad 1 \le x \le e$$

Hence,

$$E[eX]=E[Y]=\int_{-\infty}^{\infty} xf_Y(x)\,dx=\int_1^e 1\,dx=e-1$$

Although the method employed in **Example 2b** to compute the expected value of a function of X is always applicable, there is, as in the discrete case, an alternative way of proceeding. The following is a direct analog of **Proposition 4.1** of **Chapter 4**.

Proposition 2.1

If X is a continuous random variable with probability density function f(x), then, for any real-valued

function g,

$$E[g(X)]=\int_{-\infty}^{\infty} g(x)f(x)\,dx$$

An application of **Proposition 2.1** to **Example 2b** yields

$$E[eX]=\int_0^1 ex\,dx \qquad \text{since } f(x)=1,\quad 0<x<1=e-1$$

which is in accord with the result obtained in that example.

The proof of **Proposition 2.1** is more involved than that of its discrete random variable analog. We will present such a proof under the provision that the random variable g(X) is nonnegative. (The general proof, which follows the argument in the case we present, is indicated in **Theoretical Exercises 5.2** and **5.3**.) We will need the following lemma, which is of independent interest.

Lemma 2.1

For a nonnegative random variable Y,

$$E[Y]=\int_0^{\infty} P\{Y>y\}\,dy$$

Proof We present a proof when Y is a continuous random variable with probability density function fY. We have

$$\int_0^{\infty} P\{Y>y\}\,dy=\int_0^{\infty}\int_y^{\infty} fY(x)\,dx\,dy$$

where we have used the fact that $P\{Y>y\}=\int_y^{\infty} fY(x)\,dx$. Interchanging the order of integration in the preceding equation yields

$$\int_0^{\infty} P\{Y>y\}\,dy=\int_0^{\infty}\left(\int_0^x dy\right)fY(x)\,dx=\int_0^{\infty} xfY(x)\,dx=E[Y]$$

Proof of Proposition 2.1 From **Lemma 2.1**, for any function g for which $g(x)\geq 0$,

$$E[g(X)]=\int_0^{\infty} P\{g(X)>y\}\,dy=\int_0^{\infty}\int_{x:g(x)>y} f(x)\,dx\,dy=\int_{x:g(x)>0}\int_0^{g(x)} dy\,f(x)\,dx=\int_{x:g(x)>0} g(x)f(x)\,dx$$

which completes the proof.

Example 2c

A stick of length 1 is split at a point U having density function f(u)=1, 0<u<1. Determine the expected length of the piece that contains the point p, $0\leq p\leq 1$.

Solution

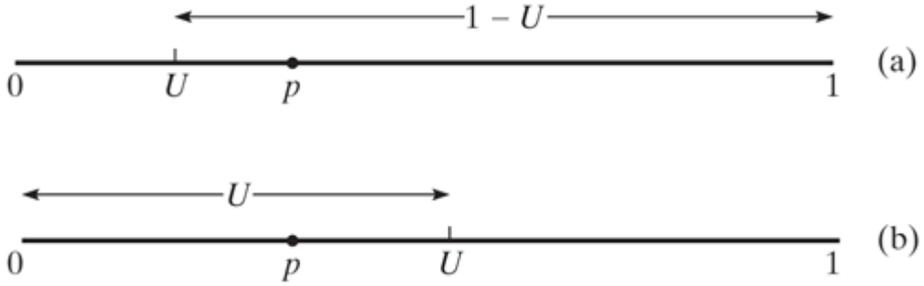
Let Lp(U) denote the length of the substick that contains the point p, and note that

$$Lp(U)=\begin{cases}1-U & U<p\\ U & U>p\end{cases}$$

(See **Figure 5.2**.) Hence, from **Proposition 2.1**,

$$E[Lp(U)]=\int_0^1 Lp(u)du=\int_0^p(1-u)du+\int_p^1 udu=12-(1-p)22+12-p22=12+p(1-p)$$

Figure 5.2 Substick containing point p: (a) U<p; (b) U>p.



Since p(1−p) is maximized when p=12, it is interesting to note that the expected length of the substick containing the point p is maximized when p is the midpoint of the original stick.

Example 2d

Suppose that if you are s minutes early for an appointment, then you incur the cost *cs*, and if you are s minutes late, then you incur the cost *ks*. Suppose also that the travel time from where you presently are to the location of your appointment is a continuous random variable having probability density function f. Determine the time at which you should depart if you want to minimize your expected cost.

Solution

Let X denote the travel time. If you leave t minutes before your appointment, then your cost—call it Ct(X)—is given by

$$Ct(X)=\begin{cases}c(t-X) & \text{if } X\le t\\ k(X-t) & \text{if } X\ge t\end{cases}$$

Therefore,

$$E[Ct(X)]=\int_0^\infty Ct(x)f(x)\,dx=\int_0^t c(t-x)f(x)\,dx+\int_t^\infty k(x-t)f(x)\,dx=ct\int_0^t f(x)\,dx-c\int_0^t xf(x)\,dx+k\int_t^\infty xf(x)\,dx-kt\int_t^\infty f(x)\,dx$$

The value of t that minimizes E[Ct(X)] can now be obtained by calculus. Differentiation yields

$$\frac{d}{dt}E[Ct(X)]=ct\,f(t)+cF(t)-ct\,f(t)-kt\,f(t)+kt\,f(t)-k[1-F(t)]=(k+c)F(t)-k$$

Equating the rightmost side to zero shows that the minimal expected cost is obtained when you leave t* minutes before your appointment, where t* satisfies

$$F(t^*)=\frac{k}{k+c}$$

As in **Chapter 4**, we can use **Proposition 2.1** to show the following.

Corollary 2.1

If a and b are constants, then

$$E[aX+b]=aE[X]+b$$

The proof of **Corollary 2.1** for a continuous random variable X is the same as the one given for a discrete random variable. The only modification is that the sum is replaced by an integral and the probability mass function by a probability density function.

The variance of a continuous random variable is defined exactly as it is for a discrete random variable, namely, if X is a random variable with expected value μ, then the variance of X is defined (for any type

of random variable) by

$$\text{Var}(X)=E[(X-\mu)2]$$

The alternative formula,

$$\text{Var}(X)=E[X2]-(E[X])2$$

is established in a manner similar to its counterpart in the discrete case.

Example 2e

Find Var(X) for X as given in **Example 2a**.

Solution

We first compute E[X2].

$$E[X2]=\int_{-\infty}^{\infty} x2f(x)\,dx=\int_{0}^{1} 2x3\,dx=12$$

Hence, since E[X]=23, we obtain

$$\text{Var}(X)=12-(23)2=118$$

It can be shown that, for constants a and b,

$$\text{Var}(aX+b)=a2\text{Var}(X)$$

The proof mimics the one given for discrete random variables.

There are several important classes of continuous random variables that appear frequently in applications of probability; the next few sections are devoted to a study of some of them.

5.3 The Uniform Random Variable

A random variable is said to be *uniformly* distributed over the interval (0, 1) if its probability density function is given by

$$f(x)=\begin{cases}1 & 0<x<1\\ 0 & \text{otherwise}\end{cases} \tag{3.1}$$

Note that **Equation (3.1)** is a density function, since $f(x)\geq 0$ and $\int_{-\infty}^{\infty} f(x)\,dx=\int_0^1 dx=1$. Because $f(x)>0$ only when $x\in(0,1)$, it follows that X must assume a value in interval (0, 1). Also, since f(x) is constant for $x\in(0,1)$, X is just as likely to be near any value in (0, 1) as it is to be near any other value. To verify this statement, note that for any $0<a<b<1$,

$$P\{a\leq X\leq b\}=\int_a^b f(x)\,dx=b-a$$

In other words, the probability that X is in any particular subinterval of (0, 1) equals the length of that subinterval.

In general, we say that X is a uniform random variable on the interval (α, β) if the probability density function of X is given by

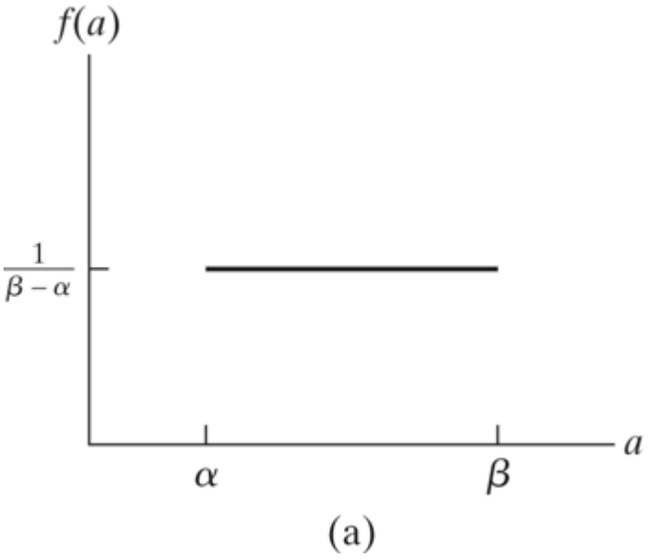
$$f(x)=\begin{cases}\frac{1}{\beta-\alpha} & \text{if } \alpha<x<\beta\\ 0 & \text{otherwise}\end{cases} \tag{3.2}$$

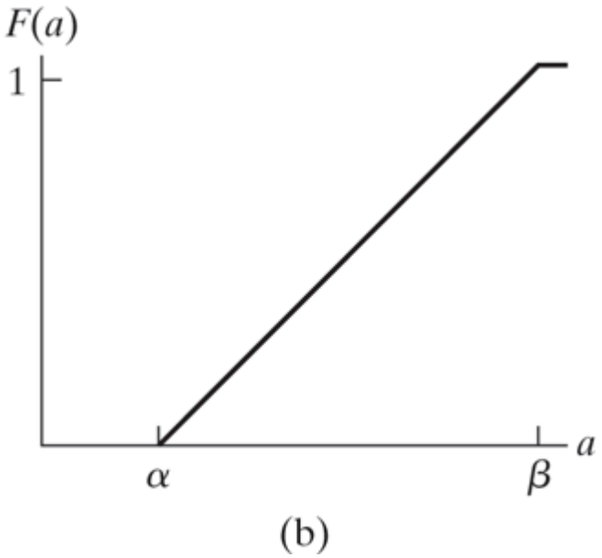
Since $F(a)=\int_{-\infty}^{a} f(x)\,dx$, it follows from **Equation (3.2)** that the distribution function of a uniform random variable on the interval (α, β) is given by

$$F(a)=\begin{cases}0 & a\leq\alpha\\ \frac{a-\alpha}{\beta-\alpha} & \alpha<a<\beta\\ 1 & a\geq\beta\end{cases}$$

Figure 5.3 presents a graph of f(a) and F(a).

Figure 5.3 Graph of (a) f(a) and (b) F(a) for a uniform (α, β) random variable.





(b)

Example 3a

Let X be uniformly distributed over (α, β). Find (a) E[X] and (b) Var(X).

Solution

a. $E[X]=\int_{-\infty}^{\infty} xf(x)\,dx=\int_{\alpha}^{\beta} x\beta-\alpha dx=\beta^2-\alpha^2 2(\beta-\alpha)=\beta+\alpha 2$

   In words, the expected value of a random variable that is uniformly distributed over some interval is equal to the midpoint of that interval.
b. To find Var(X), we first calculate E[X2].
   $E[X2]=\int_{\alpha}^{\beta} 1\beta-\alpha x2\,dx=\beta^3-\alpha^3 3(\beta-\alpha)=\beta^2+\alpha\beta+\alpha^2 3$

   Hence,
   $Var(X)=\beta^2+\alpha\beta+\alpha^2 3-(\alpha+\beta)^2 4=(\beta-\alpha)^2 12$

   Therefore, the variance of a random variable that is uniformly distributed over some interval is the square of the length of that interval divided by 12.

Example 3b

If X is uniformly distributed over (0, 10), calculate the probability that (a) X<3, (b) X>6, and (c) 3<X<8.

Solution

a. $P\{X<3\}=\int_0^3 110\,dx=310$
b. $P\{X>6\}=\int_6^{10} 110\,dx=410$
c. $P\{3<X<8\}=\int_3^8 110\,dx=12$

Example 3c

Buses arrive at a specified stop at 15-minute intervals starting at 7 A.M. That is, they arrive at 7, 7:15, 7:30, 7:45, and so on. If a passenger arrives at the stop at a time that is uniformly distributed between 7 and 7:30, find the probability that he waits

a. less than 5 minutes for a bus;
b. more than 10 minutes for a bus.

Solution

Let X denote the number of minutes past 7 that the passenger arrives at the stop. Since X is a uniform random variable over the interval (0, 30), it follows that the passenger will have to wait less than 5 minutes if (and only if) he arrives between 7:10 and 7:15 or between 7:25 and 7:30. Hence, the desired probability for part (a) is

$$P\{10<X<15\}+P\{25<X<30\}=\int_{10}^{15}\frac{1}{30}dx+\int_{25}^{30}\frac{1}{30}dx=\frac{1}{3}$$

Similarly, he would have to wait more than 10 minutes if he arrives between 7 and 7:05 or between 7:15 and 7:20, so the probability for part (b) is

$$P\{0<X<5\}+P\{15<X<20\}=\frac{1}{3}$$

The next example was first considered by the French mathematician Joseph L. F. Bertrand in 1889 and is often referred to as *Bertrand's paradox*. It represents our initial introduction to a subject commonly referred to as *geometrical probability*.

Example 3d

Consider a random chord of a circle. What is the probability that the length of the chord will be greater than the side of the equilateral triangle inscribed in that circle?

Solution

As stated, the problem is incapable of solution because it is not clear what is meant by a random chord. To give meaning to this phrase, we shall reformulate the problem in two distinct ways.

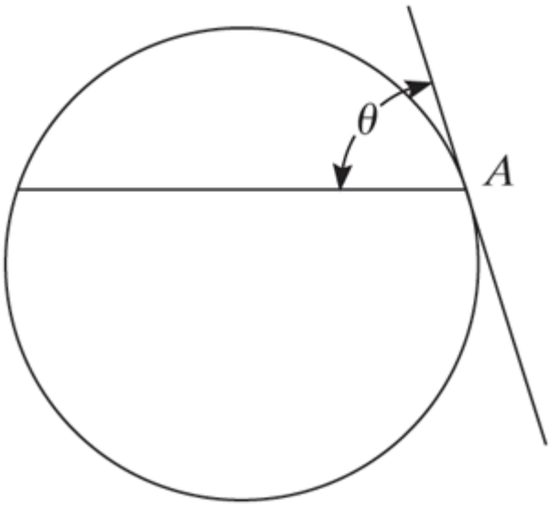
The first formulation is as follows: The position of the chord can be determined by its distance from the center of the circle. This distance can vary between 0 and r, the radius of the circle. Now, the length of the chord will be greater than the side of the equilateral triangle inscribed in the circle if the distance from the chord to the center of the circle is less than r/2. Hence, by assuming that a random chord is a chord whose distance D from the center of the circle is uniformly distributed between 0 and r, we see that the probability that the length of the chord is greater than the side of an inscribed equilateral triangle is

$$P\{D<\frac{r}{2}\}=\frac{r/2}{r}=\frac{1}{2}$$

For our second formulation of the problem, consider an arbitrary chord of the circle; through one end of the chord, draw a tangent. The angle θ between the chord and the tangent, which can vary from 0◦ to 180◦, determines the position of the chord. (See **Figure 5.4**.) Furthermore, the length of the chord will be greater than the side of the inscribed equilateral triangle if the angle θ is between 60◦ and 120◦. Hence, assuming that a random chord is a chord whose angle θ is uniformly distributed between 0◦ and 180◦, we see that the desired answer in this formulation is

$$P\{60<\theta<120\}=\frac{120-60}{180}=\frac{1}{3}$$

Figure 5.4



Note that random experiments could be performed in such a way that 12 or 13 would be the correct probability. For instance, if a circular disk of radius r is thrown on a table ruled with parallel lines a distance 2r apart, then one and only one of these lines would cross the disk and form a chord. All distances from this chord to the center of the disk would be equally likely, so that the desired probability that the chord's length will be greater than the side of an inscribed equilateral triangle is 12. In contrast, if the experiment consisted of rotating a needle freely about a point A on the edge (see **Figure 5.4**) of the circle, the desired answer would be 13.